MILITARY SPECIFICATION

MODULES, DIGITAL, STANDARD ELECTRONIC

68000 SUPPORT MODULE, KEY CODE ASC

This amendment forms a part of Military Specification MIL-M-28787/351, dated 26 October 1990, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 6

TABLE I, Test condition H44, Max column: Delete "-0.5" (three places) and substitute "-0.2" (three places).

PAGE 8

TABLE I, under test condition Z at the bottom of the page, add the following:

"

AA	Noise immunity	XXVII	Q	------See table XXVII-----

"

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PAGE 13

4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through AA shall be performed in any order with the following exceptions. Test conditions A, C, D, E and F are stress tests and shall be performed only at 25 ± 5°C. Test condition AA is a design verification test and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or F are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E and F where applicable, followed by the remaining tests in the sequence."

4.5.1: Delete and substitute the following:

"4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W and AA in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B and G through U shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown for operating temperature tests in the test requirements document."

PAGE 36

Note 1, Isolation test tolerance table, test condition column: Delete "A1-A2" and substitute "A1".

Note 1, Isolation test tolerance table, test condition column: Delete "A3-A16" and substitute "A2-A16".

PAGE 60

TABLE IX, Test condition H44, Max column: Delete "-0.5" (one place) and substitute "-0.2" (one place).

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PAGE 144

TABLE XXVII: Add a new table XXVII as printed on page 4 of this amendment.

PAGES 145 THROUGH 149

APPENDIX: Delete old appendix, pages 144 through 148, and substitute new appendix, pages 145 through 149, as printed on pages 5 through 9 of this amendment.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Preparing activity:
Navy - NW

(Project 5963-0270)

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

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TABLE XXVII. Noise immunity test.

Test cond	Parameter	Notes	Fig.
AA	Noise immunity	1	N/A

NOTES:

1. The actual test methodology is design dependent. This requirement might be verified by module test, component test or design analysis. Pass/fail will be determined by the SEM-QAA, Code 603, Naval Weapons Support Center Crane as part of the module qualification process.

2. This requirement verifies proper operation of the DTACK CONTROL LOGIC. Paragraph 30 of the appendix contains a description of the operation of the DTACK CONTROL LOGIC.

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APPENDIX

MODULE FUNCTION

10. SCOPE

10.1 Scope. This appendix contains information regarding the module function. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. Module function. The ASC module provides dedicated support functions to a 68000 based microprocessor being operated at up to 6 MHz. The ASC module contains DTACK and data strobe support logic. A PAL is used to generate complex signals DS EN and DTACK EN. The boolean equations for these outputs are on figure 1.

$$
\begin{aligned}
\text{DS EN} = {} & \overline{\text{PROM3}} \cdot \overline{\text{PROM2}} + \overline{\text{PROM3}} \cdot \text{DELAY SELEN} \cdot \text{BES} + \text{PROM3} \cdot \overline{\text{PROM1}} \cdot \text{DELAY SELEN} \cdot \text{BES} \\
& + \text{PROM3} \cdot \text{PROM1} \cdot (\text{DELAY SEL2} \cdot \text{DELAY SEL1} \cdot \text{DELAY SEL0} + \overline{\text{DELAY SEL2}} \cdot \overline{\text{DELAY SEL1}} \cdot \overline{\text{DELAY SEL0}}) \\
& + \text{PROM3} \cdot \text{PROM1} \cdot \text{DELAY SELEN} (\text{BES} \cdot \overline{\text{DELAY SEL2}} + \text{BES} \cdot \text{DELAY SEL2} \cdot \overline{\text{DELAY SEL1}} \cdot \overline{\text{DELAY SEL0}} + \text{CNT1} \cdot \text{DELAY SEL2} \cdot \text{DELAY SEL1} \cdot \text{DELAY SEL0}) \\
\text{DTACK EN} = {} & \overline{\text{PROM3}} \cdot \overline{\text{-XDS}} \cdot \text{BES} + \text{PROM3} \cdot \text{PROM1} (\overline{\text{DELAY SEL2}} \cdot \overline{\text{DELAY SEL1}} \cdot \overline{\text{DELAY SEL0}} \\
& + \text{DELAY SEL2} \cdot \text{DELAY SEL1} \cdot \text{DELAY SEL0}) + \text{PROM3} \cdot \overline{\text{PROM1}} \cdot \text{CNT1} \cdot \text{CNT0} \\
& + \text{PROM3} \cdot \text{PROM1} (\overline{\text{DELAY SEL2}} \cdot \overline{\text{DELAY SEL1}} \cdot \text{DELAY SEL0} \cdot \text{CNT1} + \text{DELAY SEL2} \cdot \overline{\text{DELAY SEL1}} \cdot \text{CNT2} \cdot \text{CNT0} + \overline{\text{DELAY SEL2}} \cdot \text{DELAY SEL1} \cdot \overline{\text{DELAY SEL0}} \cdot \text{CNT2} \\
& + \overline{\text{DELAY SEL2}} \cdot \text{DELAY SEL1} \cdot \text{DELAY SEL0} \cdot \text{CNT3} \cdot \text{CNT2} \cdot \text{CNT0})
\end{aligned}
$$

FIGURE 1. ASC PAL boolean equations.

An "external DTACK" multiplexer is used to select the DTACK input pin. The "external DTACK multiplexer output signal", -XDS, pin 74, is the active low output of the external DTACK multiplexer circuit and is synchronized to the -6 MHz C clock input, pin 12. The -XDS signal, pin 74, is intended primarily for module testing and has no functional use. However, -XDS is also useful as an indication of proper operation of the XDS flip-flop which sources the -XDS output signal. With the module input conditions defined in figures 9h, 9i and 9j of MIL-M-28787/351, the switching of the -XDS output is

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synchronized to the falling edge of the -6 MHz C clock. These input conditions provide adequate set-up and hold times assuring the proper operation of the XDS flip-flop. Under certain conditions, the inputs defined in INPUT PATTERNS A, B, C and D of figures 9h, 9i and 9j can be asynchronous causing the set-up and hold times on the XDS flip-flop to be violated. It is essential that the XDS flip-flop properly respond to the asynchronous inputs at the first or very next -6 MHz C falling edge. When flip-flop devices from certain manufacturers are used for the XDS flip-flop, asynchronous inputs were found to cause the -XDS output to enter a metastable state (output oscillation) resulting in a delayed response to the inputs of one, two or more -6 MHz C clock edges. When a binary code is placed on inputs PROM0-PROM2 (LSD to MSB), the corresponding DTACK input (-DTACK MUX 0 - -DTACK MUX 7) is selected and synchronized to the rising edge of 6 MHz B. The synchronized output (-XDS) becomes active when the selected input goes low. PROM0-PROM3 are encoded inputs that give the module information about the peripheral being accessed. The state of PROM3, for example, tells the ASC module whether the DTACK signal will be generated by the module being accessed, or generated locally on the ASC module. When PROM3 is low, the DTACK signal must be generated externally, and PROM0-PROM3 select the input pin for the awaited DTACK signal (see table I). When PROM3 is high, the DTACK signal is generated locally by decoding other PAL inputs. The number of wait states to be inserted before a locally generated DTACK signal is determined by the DELAY SEL inputs (DELAY SEL 0-DELAY SEL 2), the CNT inputs (CNT0-CNT3), and PROM1. Table I shows the number of wait states for various input combinations when the general purpose counter outputs (Q0-Q3) are connected to CNT0-CNT3, respectively, and the counter is running at the same clock frequency as the 68000. The number of wait states can be altered by changing the clock frequency of the counter, or by preloading the counter with a number other than 0. The module contains a remote request latch and a READ/-WRITE latch. When PROM2 and PROM1 are high, the remote request output (-REMREQ) becomes active on the rising edge of the remote request input (REM). -REMREQ remains active until -AS or -UDS, -LDS, and READ go high. READL1 and READL2 are outputs of a transparent latch. When ADDL goes high, READL1 and READL2 will be latched low until ADDL goes low again. A quadruple 2 input multiplexer is fully pinned out for general purpose use. When SELB is low, inputs A0-A3 correspond to outputs Y0-Y3, respectively, and when SELB is high, B0-B3 correspond to Y0-Y3. -OE is the output enable signal. When -OE is low, Y0-Y3 are enabled. When -OE is high, Y0-Y3 are in tristate. A four bit synchronous binary counter is pinned out

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TABLE I. Truth table of DTACK EN logic.

PROM Inputs				DELAY SEL inputs			DTACK source	Number of wait states*
3	2	1	0	2	1	0		
0	0	0	0	X	X	X	-DTACK MUX 0 Input	Determined by DTACK source
0	0	0	1	X	X	X	-DTACK MUX 1 Input	Determined by DTACK source
0	0	1	0	X	X	X	-DTACK MUX 2 Input	Determined by DTACK source
0	0	1	1	X	X	X	-DTACK MUX 3 Input	Determined by DTACK source
0	1	0	0	X	X	X	-DTACK MUX 4 Input	Determined by DTACK source
0	1	0	1	X	X	X	-DTACK MUX 5 Input	Determined by DTACK source
0	1	1	0	X	X	X	-DTACK MUX 6 Input	Determined by DTACK source
0	1	1	1	X	X	X	-DTACK MUX 7 Input	Determined by DTACK source
1	X	0	X	X	X	X	Internal ASC Generated	3
1	X	1	X	0	0	0	Internal ASC Generated	0
1	X	1	X	0	0	1	Internal ASC Generated	2
1	X	1	X	0	1	0	Internal ASC Generated	4
1	X	1	X	0	1	1	Internal ASC Generated	13
1	X	1	X	1	0	0	Internal ASC Generated	5
1	X	1	X	1	0	1	Internal ASC Generated	5
1	X	1	X	1	1	0	NOT USED	
1	X	1	X	1	1	1	Internal ASC Generated	0

* Number of wait states assumes the Q0-Q3 are connected to CNT0-CNT3, that the counter is operating at the same frequency as the 68000 MPU, and that the counter is preloaded with 0000.

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for general purpose use. Parallel loading is accomplished by placing data on D0-D3 and driving -LD EN low. On the next rising edge of the clock (CLK), the input data is preloaded into the counter. When CNT EN, -RST, and -LD EN are high, the binary outputs (Q0-Q3) synchronously increment on each rising edge of the clock. When all four Q outputs are high, TC will go high. CNT EN is a count enable input. When CNT EN is low, counting is stopped and TC is disabled. The asynchronous reset input (-RST) clears all outputs (Q0-Q3). The ASC module generates bus control strobes which support both 68000 microprocessor and DMA controlled data transfers. When -BEN is active, it indicates that the 68000 is controlling the address and data buses, and that input signals (-AS, -UDS, -LDS, and READ) are synchronous to the clock (6 MHZ B). When -BEN is inactive, it indicates that a DMA controller or other supervisor is performing a transfer. In such a case, -AS, -UDS, -LDS, and READ are assumed to be asynchronous. The SOCS signifies the start of a bus cycle when -AS, and -UDS, -LDS or READ go low. On the next rising edge of the clock (6 MHZ B), BES and -BES become active. The bus control strobe generation logic then waits until the DS EN inputs (PROM3, PROM2, BES, DELAY SELEN, DELAY SEL 0, DELAY SEL 1, and DELAY SEL 2) meet the conditions required to generate DS EN. Once DS EN becomes active, the next rising edge of the clock enables the conditioned data strobes. -DSW goes low if READ is low, -CUDS goes low if -UDS is low, and -CLDS goes low if -LDS is low. At this point, the circuit waits for DTACK EN to be generated. On the rising edge of the clock after DTACK EN is generated, DTACK and -DTACK become active. On the rising edge of the clock following DTACK generation, ADDL and -ADDL become active. When -AS or -UDS, -LDS and READ are negated by the bus master, SOCS, DTACK, -DTACK, -DSW, -CUDS and -CLDS are returned to their inactive states. If ADDL has been active for at least one clock period before the bus master signals were negated, then BES and -BES are also negated. If the bus master strobes are negated before ADDL has been active for at least one clock period, then BES and -BES remain active until the next rising edge of the clock. ADDL and -ADDL are negated after the rising edge of the clock following BES negation. Figure 2 shows the timing of a typical ASC bus cycle.

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APPENDIX

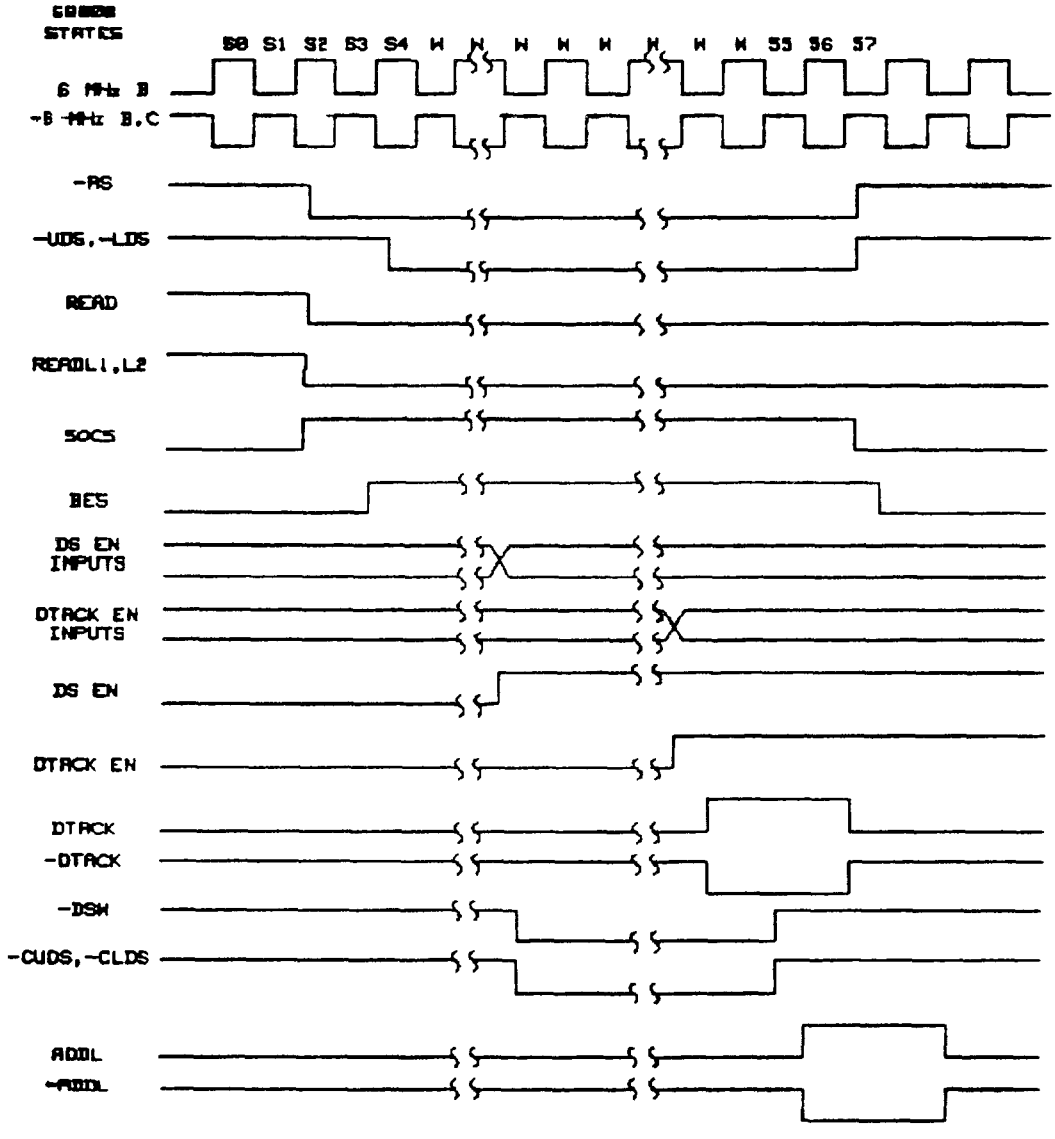


FIGURE 2. Typical ASC timing for a 68000 write cycle.

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